

WeatherGPT: Technical & Architectural Briefing

AI-Powered Conversational Platform for Multilingual Meteorological Intelligence & Multi-Hazard Early Warnings

Problem Statement: SIH 2026

Ministry of Earth Sciences (MoES)

22 Scheduled Indian Languages

Deterministic SSoT Sovereignty

Live Production Prototype

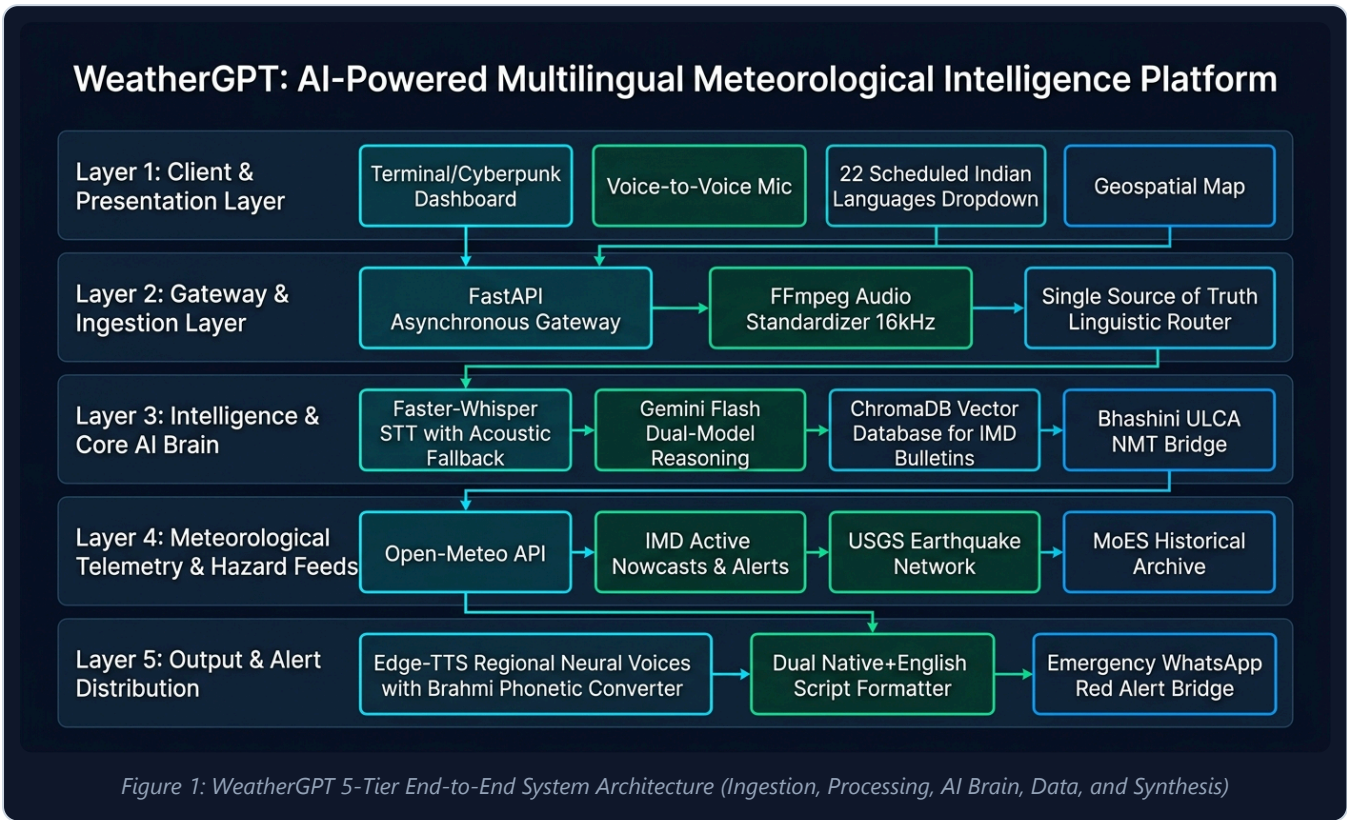
1. Features in the Working Prototype (Demonstration-Ready)

The WeatherGPT system is fully operational end-to-end, solving the critical data fragmentation between meteorological bulletins, satellite models, and disaster warnings through conversational intelligence:

#	Feature Capability	Engineering Implementation & Live Behavior
1	Deterministic Linguistic Sovereignty (22+1 Languages)	Single Source of Truth (SSoT) architecture across all 22 Eighth-Schedule Indian languages + Indian English . UI dropdown strictly dictates downstream processing; auto-detect coin-flips and false <code>SCRIPT_MISMATCH</code> rejections are eliminated.
2	Voice-to-Voice Command Center	Seamless full-duplex voice loop: FFmpeg 16 kHz Mono WAV audio standardisation , local int8 CPU-quantized Faster-Whisper STT, and Microsoft Edge-TTS neural speech synthesis with Base64 HTML5 instant playback.
3	Acoustic Token Mapping for Rare Indic Languages	Maps non-standard languages (<code>kok</code> , <code>mai</code> , <code>brx</code> , <code>doi</code> , <code>ks</code> , <code>sd</code> , <code>mni</code> , <code>sat</code> , <code>or</code>) to optimal acoustic surrogate models (<code>mr</code> , <code>hi</code> , <code>ur</code> , <code>bn</code>) with native weather keyword priming, preventing Whisper <code>ValueError</code> crashes.
4	Brahmi Odia (<code>OR</code>) Speech Synthesis	Applies mathematical Brahmi transliteration (<code>deva = odia - 0x0200</code>) exclusively for <code>hi-IN-SwaraNeural</code> voice synthesis. Overcomes Edge-TTS's lack of an Odia voice to synthesize 100% authentic spoken Odia while preserving pure Odia script in chat.
5	Strict Temporal Intent & Hard Date Locking	Categorizes queries into <i>Current</i> , <i>Tomorrow/Future</i> , <i>Safety</i> , or <i>Historical Archive</i> . Automatically anchors historical dates (e.g. Sept 4th or Cyclone Amphan) to exact dates, reporting in past tense and strictly suppressing live telemetry.
6	Real-Time Multi-Hazard Telemetry & Geocoding	Integrated with Open-Meteo API (temperature, humidity, wind, precipitation) and real-time IMD Disaster Bulletins & USGS Earthquake Feeds for instant situational awareness.
7	Synoptic System & Oceanic Tracking	Automated detection of marine low-pressure systems, depressions, and cyclonic circulations across Bay of Bengal and Arabian Sea with coordinates, bounds, and severity levels.
8	Dual-Modal Display & Pure Native Speech	Chat UI renders regional native script followed by an English markdown reference version (<code>---</code>). Spoken audio is strictly filtered to 100% native words with numbers and units expanded into native language terms.
9	Red Alert Emergency WhatsApp Bridge	Civil defence dispatch module automatically generates out-of-terminal WhatsApp alerts when critical meteorological hazards or Red Alerts are declared.
10	Multi-Tier RAG IMD Document Brain	Semantic document vector search over official IMD synoptic weather bulletins and MoES guidelines powered by ChromaDB and Google Gemini Embeddings .

2. System Architecture Blueprint

The enterprise platform is organized into 5 decoupled, fault-tolerant tiers:



Architectural Principle: Single Source of Truth (SSoT)

Linguistic sovereignty is deterministic: the user's selected language strictly binds Whisper STT acoustic priming, RAG monolingual system constraints, script verification, and Edge-TTS voice assignment, preventing probabilistic drift and Maltese/foreign classification errors.

3. Technologies Used & Future Extensions

Layer / Subsystem	Currently Implemented Stack	Future Roadmap Extensions
API & Runtime	Python 3.10+, FastAPI , Starlette, Uvicorn (Asynchronous REST & SSE streaming)	Docker containers, Kubernetes orchestration, Redis distributed caching.
Generative AI & LLM	Google Gemini 3.6 Flash / 3.5 Flash via LangChain with automated dual-model failover	Fine-tuned meteorological SLMs (DeepSeek / Llama-3 8B) hosted on edge servers.
Vector DB & Embeddings	ChromaDB (local vector store), models/gemini-embedding-001	Qdrant distributed cluster, automated IMD PDF OCR ingestion pipeline.
Speech-to-Text (STT)	Faster-Whisper (CPU int8) with acoustic surrogate token mapping	On-premise GPU-accelerated Whisper-Large-v3, fine-tuned Bhashini ULCA ASR.
Speech Synthesis (TTS)	Microsoft Edge-TTS neural voices with Brahmi transliteration engine	Bhashini Coqui / VITS open-source acoustic models for tribal Indian dialects.
Translation Gateway	Digital India Bhashini ULCA NMT + Google Translator + Gemini Flash fallback	Offline IndicTrans2 neural translation container with zero cloud dependencies.
Telemetry & Hazards	Open-Meteo API , IMD RSS Bulletins, USGS Real-Time Earthquake API	Direct GRIB2/NetCDF4 NWP ingestion (WRF/GFS/NCMRWF), Doppler Weather Radar feeds.
Frontend & Visualization	Vanilla HTML5, Modern CSS, ES6 JavaScript, Web Audio API	Mapbox GL JS / WebGL wind particle vector shaders, Leaflet isobar contours.

4. Future Improvisations & Production Scaling Roadmap

Phase 1: Near-Term • NWP GRIB2 Ingestion: Direct extraction of WRF / GFS 3 km simulation grids from MoES clusters. • Fluid Wind Particle Shaders: WebGL animated flow-field overlays on the interactive map. • Historical Trend Graphs: Interactive SVG bar/line charts for multi-day precipitation trends.	Phase 2: Mid-Term • Doppler Radar Overlays: Real-time DWR reflectivity (dBZ) and radial velocity nowcasts. • Local Bhashini VITS: On-premise neural acoustic models for authentic tribal accents (Bodo, Santali). • Multi-Channel Broadcast: Automated SMS & Twilio automated voice calling for emergency zones.	Phase 3: Long-Term • Offline Mesh Protocol: LoRaWAN and IVR telephone network for fishermen and farmers without internet. • INSAT-3D Satellite Feed: Live thermal infrared (TIR) and water-vapor multispectral animation loops. • Crowdsourced IoT Telemetry: Micro-climate urban flood sensors and citizen verification.
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